AL DHAFRA HEALTH FORUM · DEVELOPING HEALTHCARE SERVICES

SAYR AI

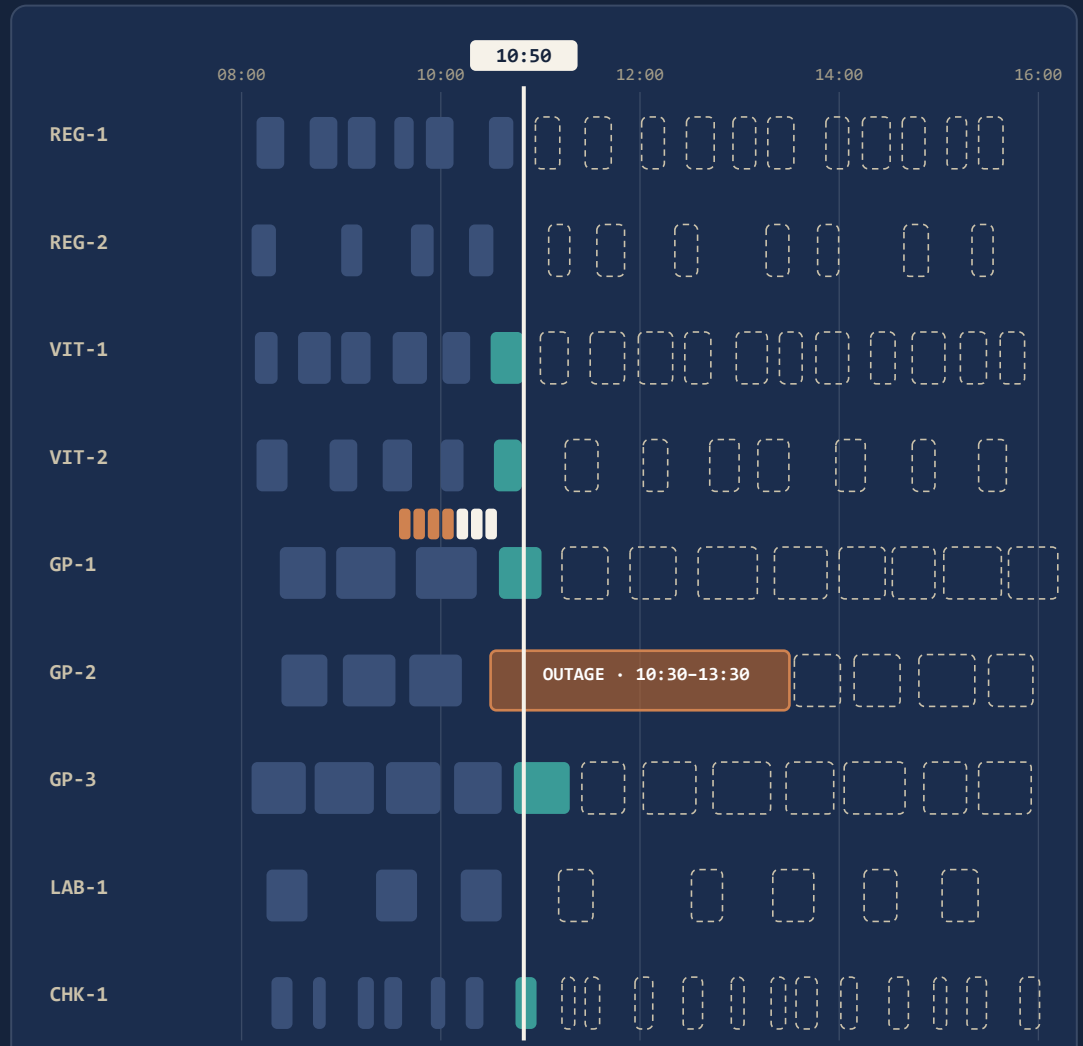
Care in Motion.

Every minute matters.
Every decision moves the system.

A simulation-driven healthcare operations decision-support prototype.
Synthetic data only.

Lujain Ahmed Mourad 8D
Precious Fuentes 8D
Eleen Amr 8D
Yosr Amgad 8D
Maryam Ammara Jelita 8D
Shaden Ahmed Mourad 6D

ADNOC Schools, Madinat Zayed



One clinician unavailable: the lane is lost, the queue grows, the plan must move.

FLOW CANVAS · ILLUSTRATION OF THE IDEA, NOT DATA

THE PROBLEM

When one delay becomes everyone's delay



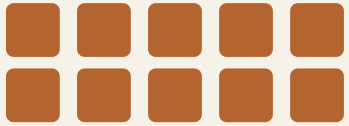
CONSULTATION QUEUE AT 11:45 · SAME PATIENTS, SAME SEED

Normal day



4 waiting
longest wait 16 min

One clinician out



10 waiting
longest wait 40 min

Healthcare delays are rarely isolated.

QUEUES GROW

11 waiting vs 5

Everyone behind the missing clinician waits. Longest wait 40 min vs 16 min at 11:45.

THE DELAY TRAVELS

37 patients

delayed 3+ min across 26 hops. Laboratory and checkout feel it later (+76 and +15 min).

JOURNEYS LENGTHEN

60 → 74 min

Average time from arrival to departure. 59 of 60 patients finish by 16:00 instead of 60.

Hypothetical outpatient journey; figures from SAYR's simulation of a fictional clinic (seed 7, staff-unavailability scenario, 10:30–13:30). No claim is made about any Al Dhafra facility.

OUR RESEARCH QUESTION

What if we could test operational decisions before they affect real patients?

SAYR is a simulation-driven healthcare operations decision-support prototype.

01 **Model a clinic** — CONTROL ROOM

02 **Simulate operations** — CONTROL ROOM

03 **Introduce a disruption** — STRESS CONTROLS

04 **Trace consequences** — RIPPLE TRACE

05 **Optimize a plan** — REFLOW STUDIO

06 **Review and approve** — ACTION DESK

07 **Measure the result** — MINUTES LEDGER

ONE DETERMINISTIC SIMULATION STATE · EVERY SCREEN READS THE SAME DATA

Synthetic clinic — seeded patients, 9 resources

› **Discrete-event simulation** — queues, services, outages, event log

› **Forecast baseline** — hourly arrivals, evaluated

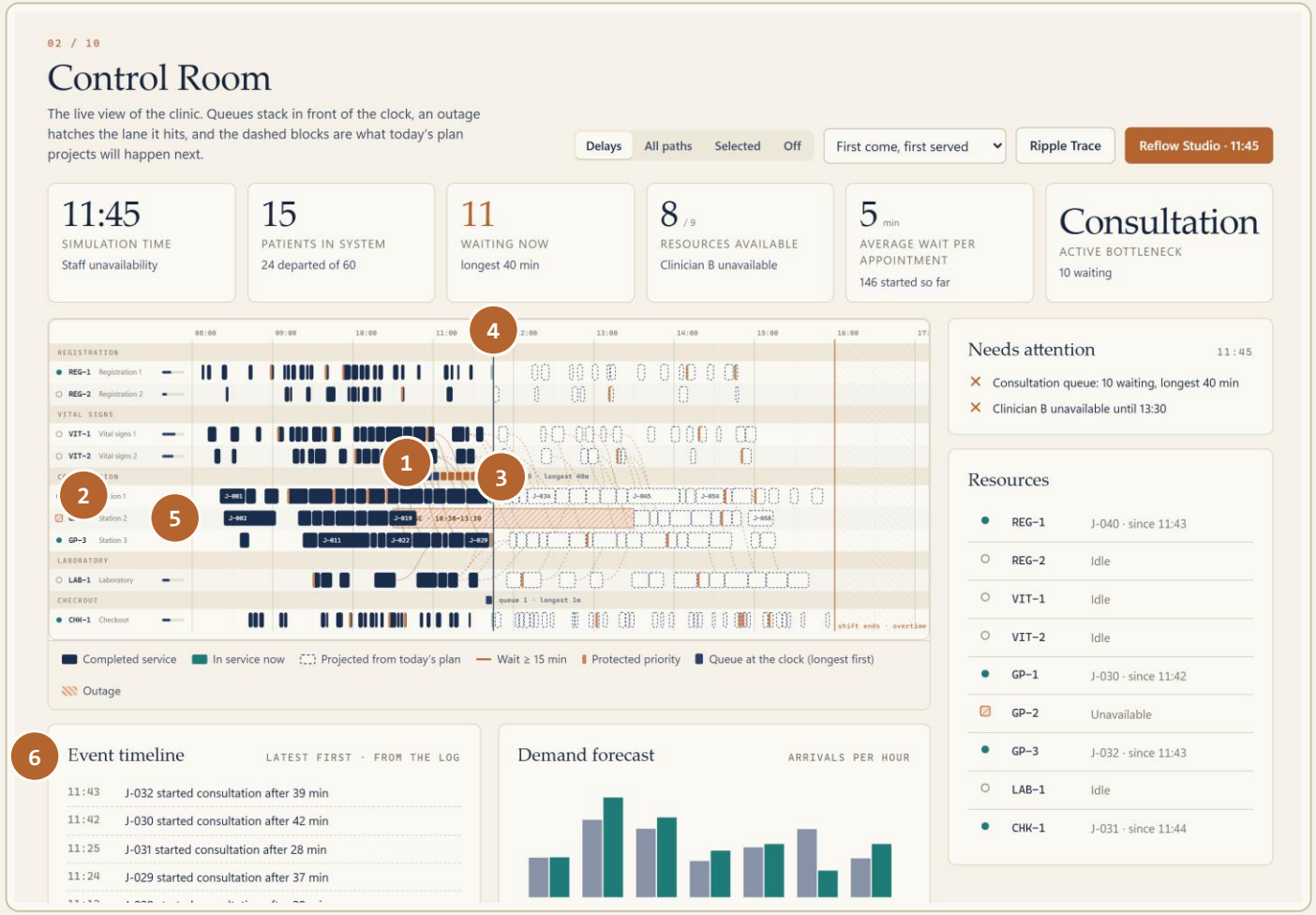
› **Reflow** — constraint-aware local search

› **Explanations** — from the event log, no language model

› **Person approves** — then the Ledger measures

No trained machine-learning model is used. Methods: simulation, a forecasting baseline evaluated against two simpler ones, and a heuristic optimizer. Source: SAYR About, methods & credits.

A digital twin of the clinic



READING THE CANVAS

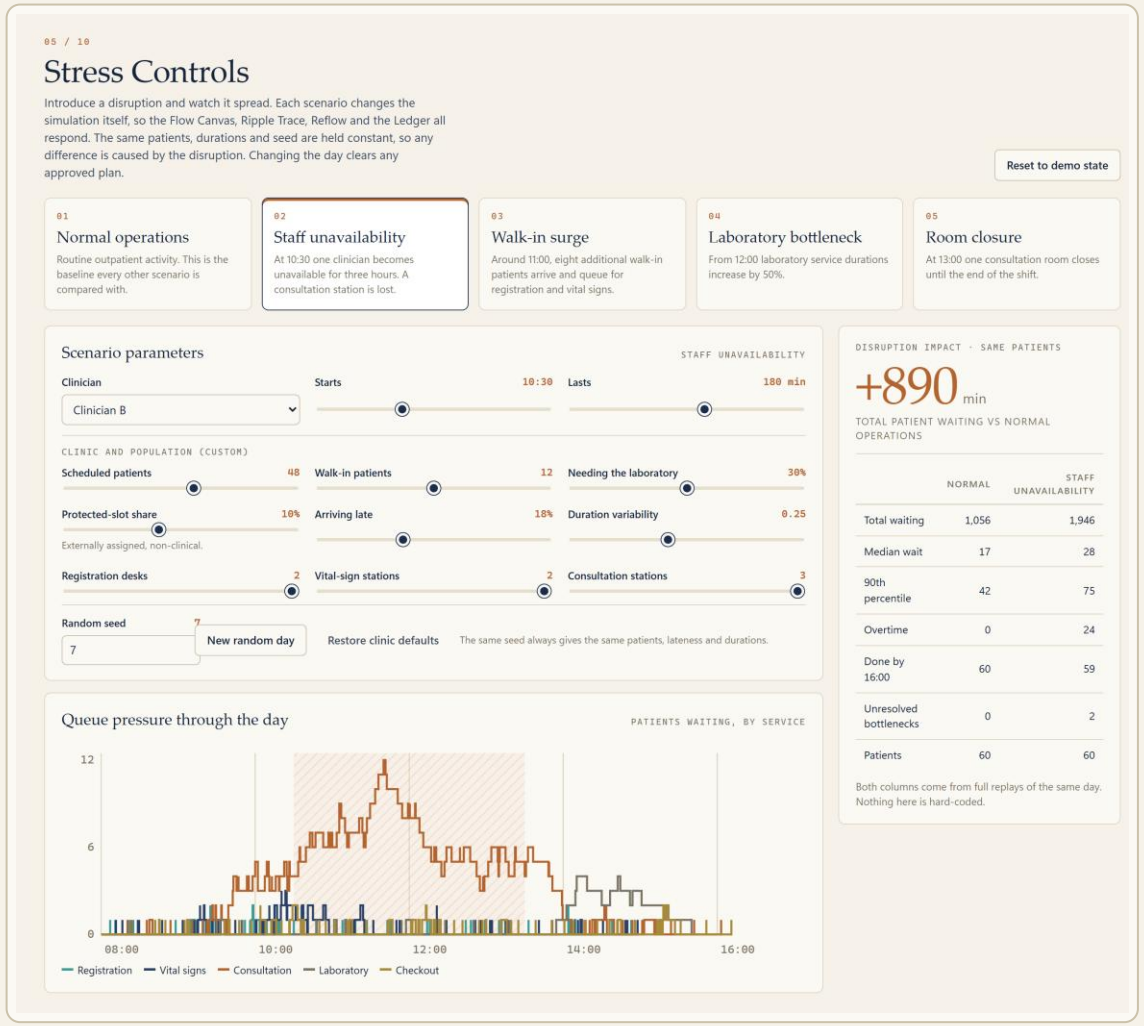
1. **Patient pathways** dotted links from a patient's stage to the next
2. **Resource lanes** nine lanes; the hatched block is the outage
3. **Queues** chips stacked in front of the clock, longest first
4. **Clock** movable; the sim recalculates from it
5. **Utilisation** small bar on each lane: busy share so far
6. **Operational events** timeline written from the event log

DISCRETE-EVENT SIMULATION, IN PLAIN WORDS

The simulator jumps from event to event (arrive, wait, start, finish, transfer, outage) and logs each one. A fixed seed regenerates the same day, so differences come from what we change.

Default clinic: 60 synthetic patients (48 scheduled, 12 walk-in); 2 registration, 2 vital-sign, 3 consultation stations (clinician + room), 1 laboratory, 1 checkout; 08:00–16:00. Shot: seed 7, staff scenario, clock 11:45.

The disruption experiment



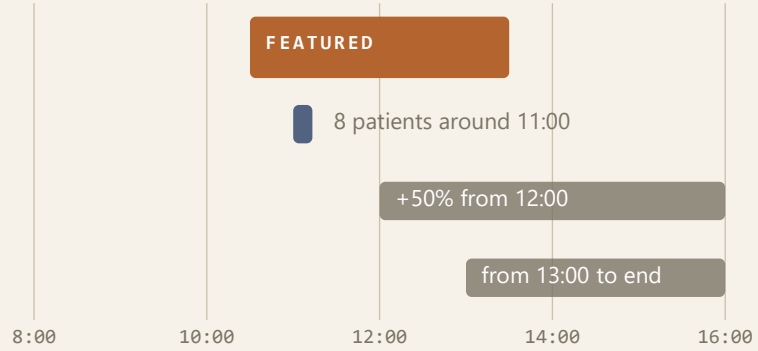
FOUR IMPLEMENTED SCENARIOS · ONE DAY, 08:00–16:00

Staff unavailability

Walk-in surge

Laboratory slowdown

Room closure

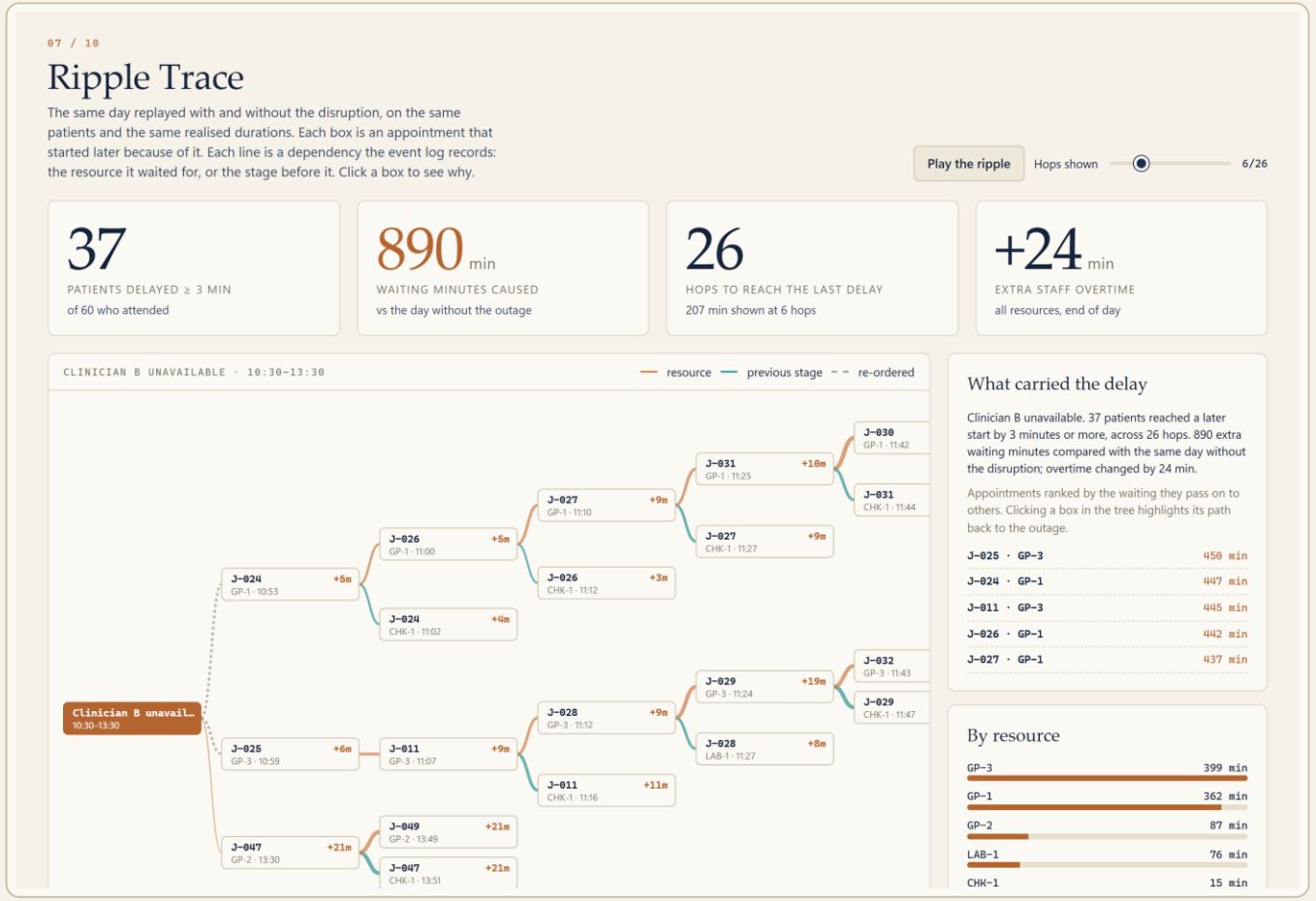


SAME 60 PATIENTS · SEED 7	NORMAL	STAFF OUT
Total waiting (min)	1,056	**1,946**
Median wait (min)	17	**28**
90th-percentile wait (min)	42	**75**
Staff overtime (min)	0	**24**
Journeys done by 16:00	60	**59**

We can investigate operational pressure without experimenting on real patients.

Source: SAYR Stress Controls, "Disruption impact · same patients", full replays of the same day. The other three scenarios are implemented but are not quantified in this deck.

Ripple Trace: following the consequences



THE CHAIN, READ FROM THE TREE · +N MIN = LATER START THAN THE SAME DAY WITHOUT THE OUTAGE

Clinician B unavailable
10:30–13:30 · station GP-2 lost

J-025 · GP-3 · 10:59
+6 min · waited for a station

J-011 · GP-3 · 11:07
+9 min · waited for GP-3

→ **J-028 · LAB-1 11:27**
+8 min · previous stage

J-028 · GP-3 · 11:12
+9 min · waited for GP-3

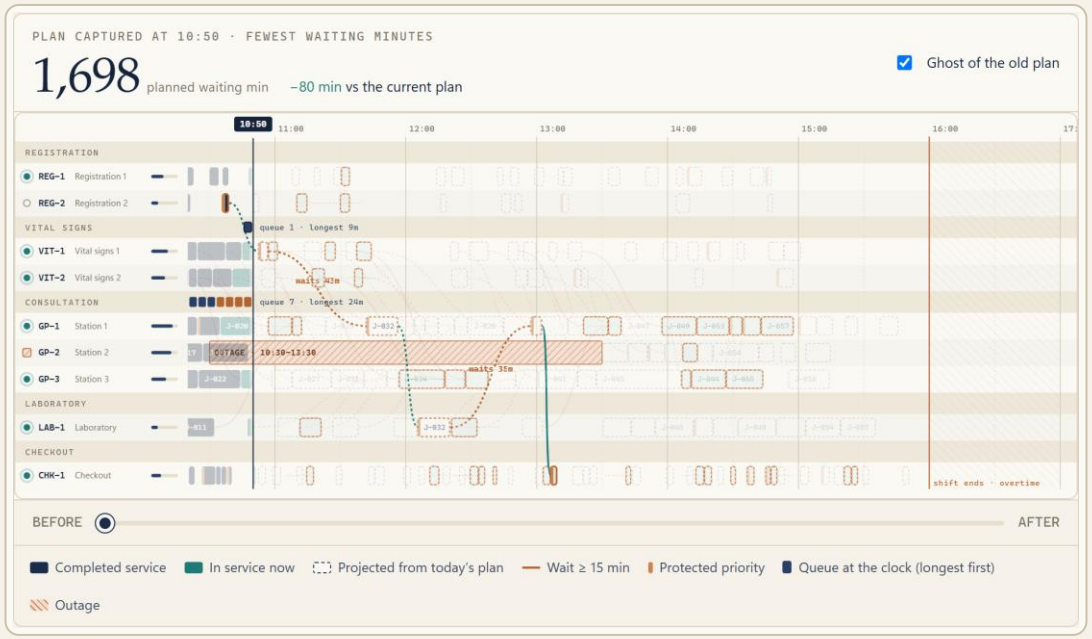
J-029 · GP-3 · 11:24
+19 min · waited for GP-3

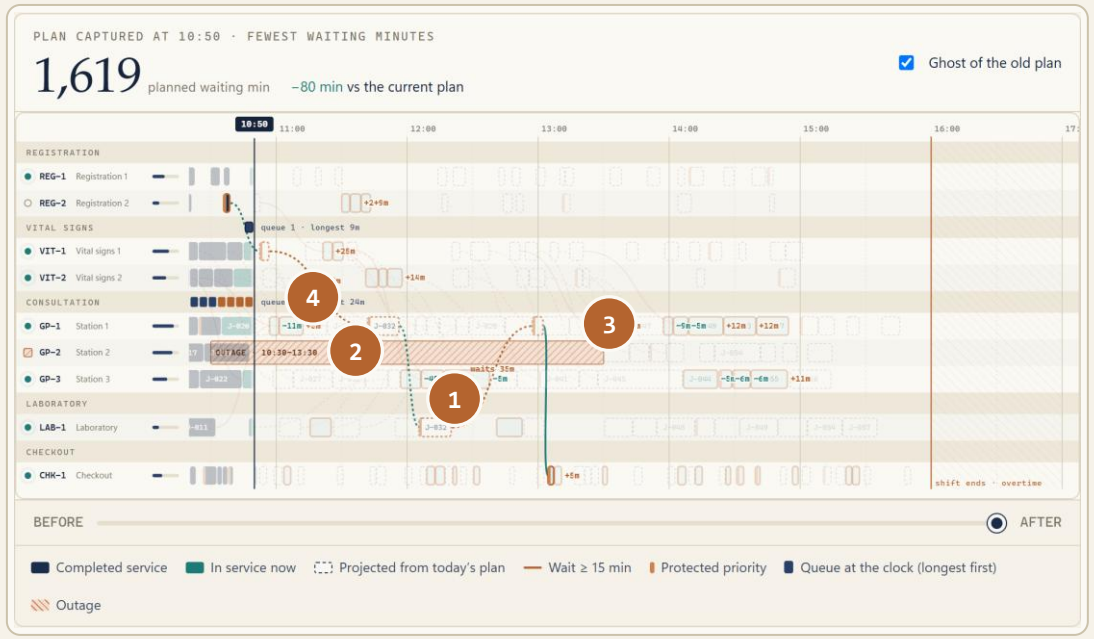
J-032 · GP-3 · 11:43
+19 min · waited for GP-3

37	890	26	+24
PATIENTS DELAYED ≥ 3 MIN	WAITING MIN CAUSED	HOPS TO THE LAST DELAY	MIN EXTRA OVERTIME

SAYR explains not just where a delay occurs, but how it spreads. Every link is a record in the event log (resource the appointment waited for, or its previous stage). Seed 7, staff scenario.

Reflow Studio: from disruption to decision





CURRENT PLAN

1,698 planned waiting min

43 appointments still expected to wait 15 min or more

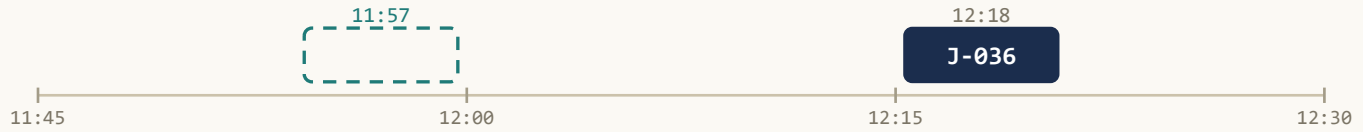
PROPOSED PLAN

1,619 planned waiting min (−80)

42 changes: 6 reassigned, 36 re-sequenced · 9 of 9 constraints kept

1 reassignment **2** outage constraint **3** re-sequenced (wait change) **4** remaining delay

ONE REAL MOVE · J-036 ON GP-3 · START 12:18 → 11:57 · −45 MIN WAIT



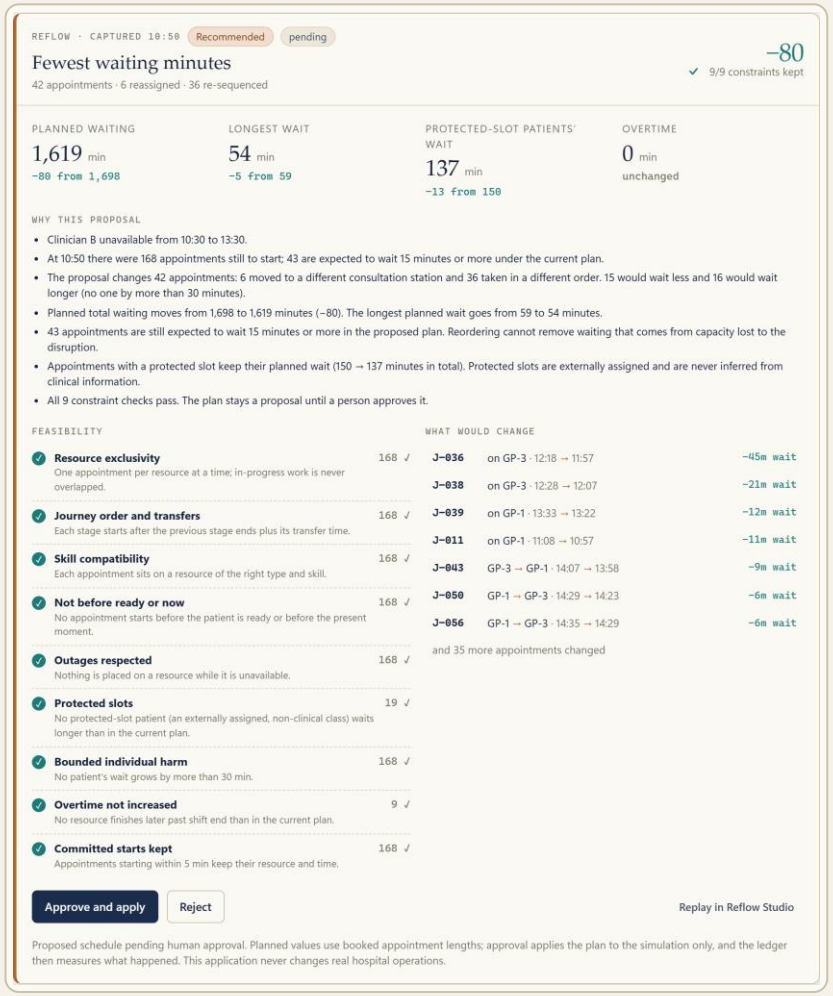
DECISION CONTROLS · NOTHING APPLIES UNTIL A PERSON CHOOSES

Accept in the simulation | Reject

Heuristic search: no guarantee of a global optimum.

Source: SAYR Reflow Studio and Action Desk, seed 7, staff scenario, plan captured at 10:50, objective "Fewest waiting minutes". Planned values use booked lengths; the Ledger measures the realised result.

The human remains in control



Reflow proposes — up to 3 alternatives → **A person reviews** — changes, constraints, trade-offs → **Approve and apply** — to the simulation only → **Audit trail** — entry is written → **Ledger measures** — what actually happened

Reject leaves today's plan untouched.

Proposed changes

42 appointments: 6 reassigned, 36 re-sequenced. The card lists each move, e.g. J-036 on GP-3, 12:18 → 11:57.

Constraints satisfied

9 of 9 checks pass, each with evidence (for example 168 appointments checked for resource exclusivity).

Trade-offs

15 appointments wait less, 16 wait longer, none by more than 30 min. 43 still expected to wait 15 min or more.

Accountability

The audit trail records each situation and decision. Protected slots are assigned externally and never inferred from clinical information.

AUDIT TRAIL (AFTER APPROVAL)

Audit trail

2 ENTRIES

1. `10:50` Reflow captured the plan at 10:50 and produced 3 alternatives, pending approval.
   ▶ Audit detail
2. `10:50` Approved "Fewest waiting minutes" (42 appointment changes). It is now the active plan.
   ▶ Audit detail

Intelligent recommendations should support accountability, not replace it.

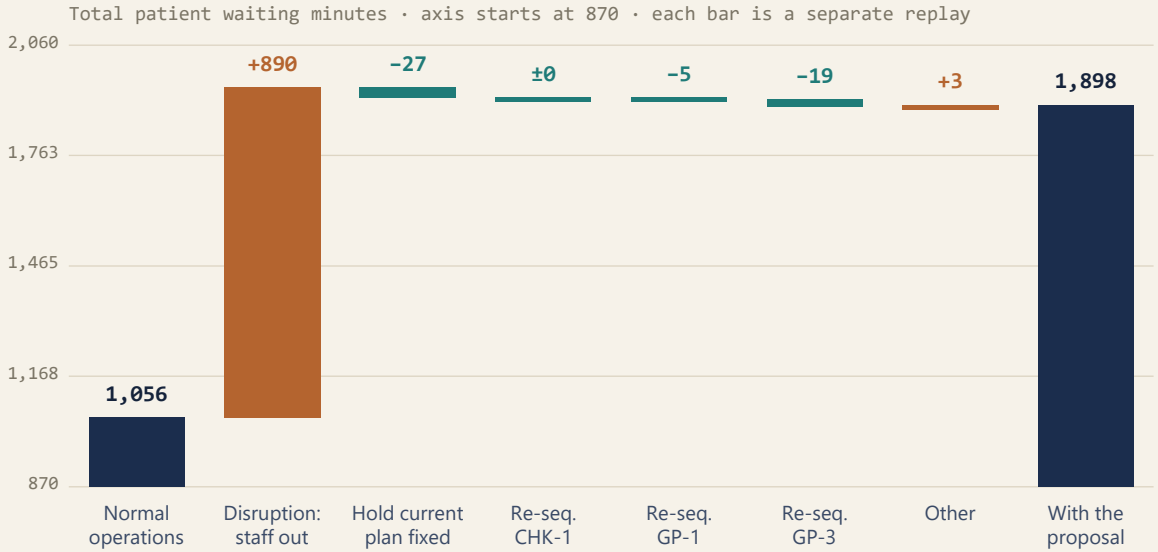
Source: SAYR Action Desk. SAYR does not autonomously determine clinical priorities. Approval applies a plan to the simulation only; the app never changes real hospital operations.

Minutes Ledger: measured, not assumed

−48 min

MEASURED CHANGE IN WAITING
−2.5% · approved plan, realised durations

−80 min WHAT THE PLANNER PROJECTED (booked lengths)



METRIC	NORMAL	DISRUPTED	OPTIMIZED	Δ
Total waiting (min)	1,056	1,946	1,898	−48
Mean wait per patient	18	32	32	−0.8
Median wait per patient	17	28	29	+1.3
90th-percentile wait	42	75	71	−3.6
Throughput by 16:00	60	59	59	±0
Incomplete at 16:00	0	1	1	±0
Resource utilisation	47%	49%	49%	±0
Staff overtime (min)	0	24	22	−1.5
Rescheduled appointments	0	0	42	+42
Constraint violations	0	0	0	0

Δ = optimized − disrupted. The 90th percentile improved while the median rose: the Ledger reports trade-offs, not one score.

COMPARISON CONDITIONS

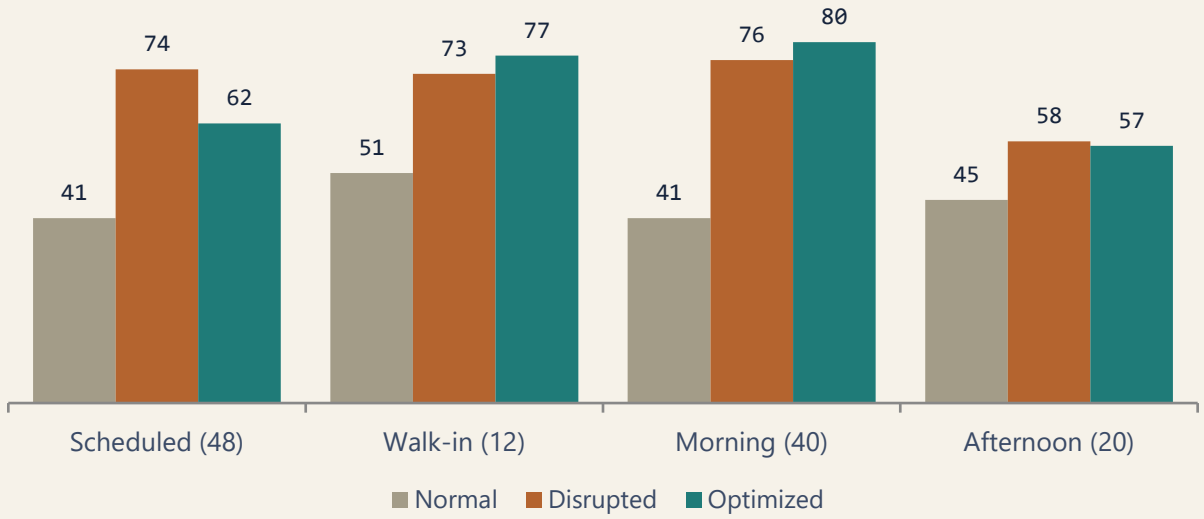
Scenario: staff unavailability (Clinician B, 10:30–13:30) · Seed 7 · 60 synthetic patients
Baseline policy: first come, first served · Reflow at 10:50, objective "fewest waiting minutes"
Same patients, arrivals and realised durations in all three runs; the bars add up exactly

SAYR evaluates whether an intervention helped, what it cost, and what trade-offs remain.

Source: SAYR Minutes Ledger (published app). Waiting = ready to start of service. Simulated, operational measures only; fewer waiting minutes is not evidence of better clinical care.

Efficiency without overlooking fairness

WAIT AT THE 90TH PERCENTILE, BY GROUP (MIN)



10 WAIT LONGER (≥ 5 min)

7 WAIT LESS (≥ 5 min)

43 ABOUT THE SAME

Optimized vs disrupted, same patients. Largest increases: J-054 +18 min, J-049 +16, J-045 (walk-in) +14, J-048 +14. Priority inversions 48 → 46.

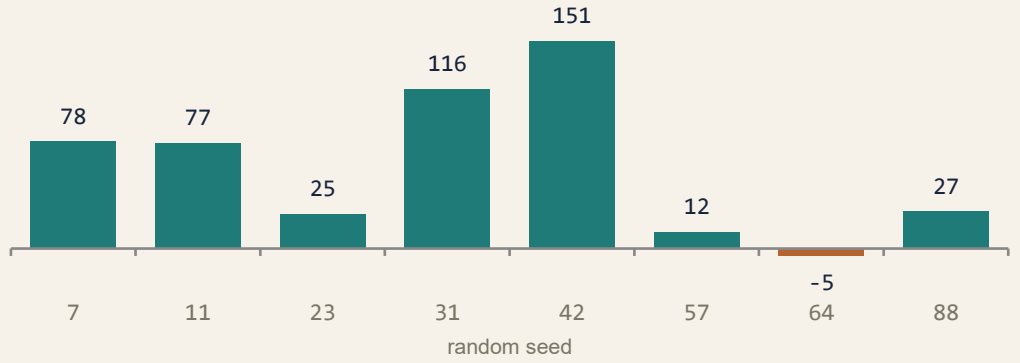
A better average does not automatically mean a fairer system.

SCHEDULING POLICIES · SAME DISRUPTED DAY

POLICY	TOTAL MIN	P90	INVERSIONS
First come, first served	1,946	75	48
Priority with aging	1,973	67	**23**
Shortest task first (aged)	**1,616**	65	45

Different objectives, different trade-offs: shortest-first needs the fewest minutes but starts a less urgent patient first 45 times.

ACROSS EIGHT SYNTHETIC DAYS · MINUTES SAVED BY REFLOW (10:45)



Helped on 7 of 8 days; worse on 1 (seed 64, +5 min). Mean saving 59.8 min, median 51.5. Nothing is cherry-picked.

Source: SAYR Minutes Ledger: fairness table, patients made worse off, scheduling policies and "Evidence across seeds". Synthetic fairness evaluation does not establish fairness in real clinical operations.

ROADMAP

From simulation to real healthcare research

PROTOTYPE TODAY → VALIDATION → SUPERVISED RESEARCH

PHASE 1 | PHASE 2 | PHASE 3 | PHASE 4 | PHASE 5

Current prototype

Synthetic patient journeys, deterministic simulation, scheduling optimization and operational evaluation.

WHERE WE ARE

Operational research collaboration

Work with healthcare operations professionals to evaluate assumptions and refine scheduling constraints.

Authorized real-world data

Investigate data access, privacy protection, data quality and integration requirements.

External validation

Compare simulator predictions with observed service patterns; assess recommendations.

Controlled deployment research

Supervised prospective testing, security assessment, workflow integration and approvals before any real use.

AL DHAFRA CONTEXT

Al Dhafra is a wide region, so coordinating services across places is a research question worth studying with local healthcare partners. We have not used, and do not state, any regional or facility performance data.

The next step is validation, not deployment by assumption.

Roadmap is a proposal for future research, not a record of work done. Real-world use would require validation with healthcare operations specialists, authorized data, and privacy and security review (SAYR, Responsible AI).

Healthcare does not operate one appointment at a time. It operates as an interconnected system, and every minute matters.

Visibility	Coordination	Explainability	Measurable outcomes	Human oversight
CONTROL ROOM	REFLOW	RIPPLE TRACE	MINUTES LEDGER	ACTION DESK

Behind every healthcare schedule is a patient waiting for care.

SAYR AI *Care in Motion*

Lujain Ahmed Mourad 8D
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ADNOC Schools, Madinat Zayed



SCAN TO TRY SAYR

maymourad.github.io/SAYR-AI-Care-In-Motion

Synthetic data only. A decision-support prototype, not a clinical tool.

Appendix A · How the simulation works

1 Next event
earliest time in the event queue

→

2 Update state
patients, resources, queues

→

3 Write to log
time, event, cause pointer

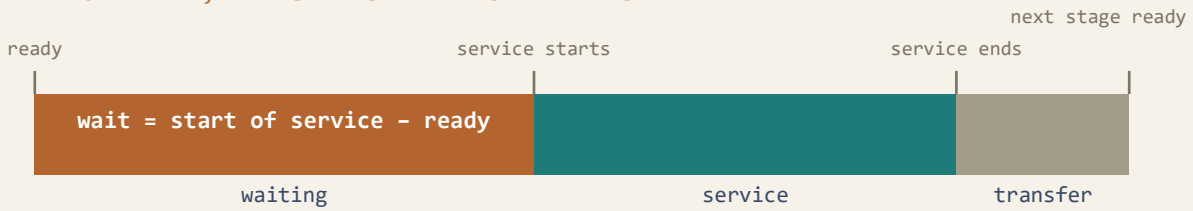
→

4 Schedule follow-ups
finish, release, transfer, next stage

EVENT TYPES

`arrival` `queue` `service start` `service finish` `release` `transfer` `outage`

ONE APPOINTMENT, TIMESTAMPS THE ENGINE KEEPS



Waiting excludes transfers and service time. Total waiting sums waits over all appointments; mean, median and 90th percentile are per patient.

Planning uses booked lengths; the replay uses realised ones (variability 0.25), which is why projected and measured savings differ.

DETERMINISM AND REPRODUCIBILITY

Seed 7 always creates the same 60 patients, lateness, no-shows and durations. Baseline, disrupted and optimized runs share the same base patients, so any difference is caused by the change.

Verified for this deck: the full demonstration path was run twice on the published app and produced identical figures (the optimizer examined 36,850 plans both times).

Source: SAYR About, methods & credits (Discrete-event simulation); Minutes Ledger methodology (seed 7). Known limits: a clinician is tied to a room; no breaks, equipment, multiple sites or clinical variation.

Appendix B · How Reflow works

SEARCH

Constraint-aware local search with seeded perturbation. It captures every appointment not yet started, tries moving it to another resource or position, keeps only legal changes and the better plan. Three runs, three objectives. A heuristic: no guarantee of a global optimum, and it may find no improvement.

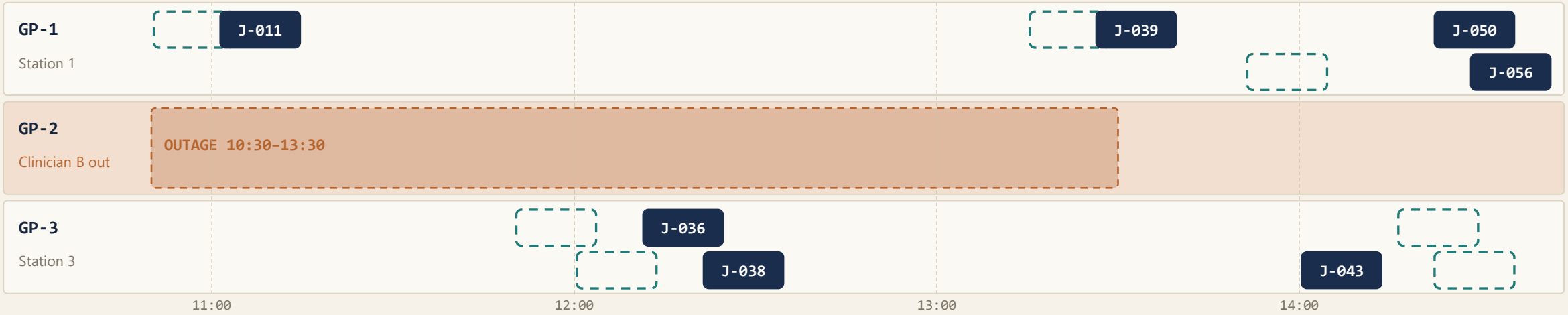
CONSTRAINTS

Hard rules: one appointment per resource; journey order and transfers; right resource type; not before ready or now; nothing on an unavailable resource; protected slots no worse; starts within 5 min kept. Limits by design: no one waits more than 30 min longer; overtime not increased.

OBJECTIVES

Fewest waiting minutes: −80 planned, 42 changes (recommended). Protect priority and the longest waits: −40, 33 changes. Fewest changes: −11, 11 changes. Different objectives give different trade-offs, which is why a person chooses.

SEVEN REAL MOVES FROM THE PROPOSAL (ACTION DESK) · CHIP = APPOINTMENT START TIME · DASHED = PROPOSED POSITION



Source: SAYR Action Desk, "What would change", 10:50 proposal, seed 7. Chips mark start times only (durations not drawn). J-043, J-050, J-056 change station.

Appendix C · Evaluation and limitations

Waiting time

Minutes between becoming ready and starting service. Transfers and service excluded. Percentages use patients who attended.

90th percentile

The wait that nine in ten patients stay under: it describes the worst-off tenth, so it can move differently from the mean.

Utilisation

Busy minutes divided by available minutes within the shift (resources that are down are not counted as available).

Fairness

Mean and 90th-percentile waits for scheduled vs walk-in and morning vs afternoon; priority inversions; protected-slot waits; patients made worse off by ≥ 5 min.

Repeated seeds

The same scenario on 8 other synthetic days, Reflow at 10:45. One run is a demonstration; the spread is the evidence.

FORECAST BASELINES · HELD-OUT ERROR (PATIENTS / HOUR)



Seasonal hourly profile chosen on 6 held-out synthetic days (14 train). Forecast 57 vs 60 actual. A baseline, not machine learning; the history is synthetic and generated from the simulator's own pattern.

LIMITS OF SIMULATED OUTCOMES

- All patients and data are synthetic; nothing is validated against a real clinic.
- Planned and measured savings differ (−80 vs −48).
- A clinician is tied to a room; no breaks, equipment, sites or clinical variation.
- Improvement is not guaranteed (1 of 8 days got worse).
- Fewer waiting minutes does not mean better care.

Source: SAYR About, methods & credits (Demand forecasting); Minutes Ledger definitions and methodology.